Problem FoL.1 ... librarian's problem

A stack of 40 books lies on the ground. Each of them is $d = 2\,\mathrm{cm}$ *thick and has mass* $m = 1\,\mathrm{kg}$
We want to place the books into 4 shelves, with each shelf containing a stack of 10 books lyin
on the side and the shelves' are at heights $100\,\mathrm{cm}$, $130\,\mathrm{cm}$, $160\,\mathrm{cm}$ *a* $190\,\mathrm{cm}$. *What work do w*
have to perform to move the books? *Mirek was unable to move full bookshelves*

Problem FoL.2 ... tough slice of bread

A square wheat field covers an area of 16 ha. *A harvesting company owns a combine harveste whose reel is* 8 m *wide. The combine moves with velocity* $5\,\mathrm{km\cdot h^{-1}}$ *and it consumes 10 litres o fuel per kilometer when harvesting. The company buys the fuel for* 28.40 *CZK per litre. How much will harvesting the whole field cost the company (rounded to integer CZK), if they hav to pay the combine driver* 85 CZK *for each started hour of work? Neglect any delays of th combine due to turning or fires in the field.*

Meggy was harvesting part-time

Problem FoL.3 ... curve

Olda is driving a car with velocity $v = 20\,\mathrm{m{\cdot}s^{-1}}$; he keeps an air freshener on his rear viev mirror. When crossing a curve, his air frshener tilted by $\alpha = 35°$. What's the radius of th curve Olda was crossing? The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Olda was driving through a curve

Problem FoL.4 ... astral

Dr Strange is an impenitent pragmatist who does not believe in out-of-body experiences. A nu
wants to correct his mistake by hitting him in the chest with a constant force and knocking th
stral body (with mass identical to that of Dr Strange, $m = 80\,\mathrm{kg}$) out of him. If the impac
(contact of the fist and the chest) took $t = 0.1\,\mathrm{s}$ and the astral body was moving with velocit
$= 5\,\mathrm{m{\cdot}s^{-1}}$ afterwards, what force did the nun hit with? The body of Dr Strange itself doe
not move, violation of mass conservation is not our problem.

Mirek was relaxing and watching trailers

Problem FoL.5 ... don't mind the tune

Meggy brought a flute to a campfire. The flute is made of the ABS polymer with therma expansion coefficient $\alpha = 9 \cdot 10^{-5}\,\mathrm{K}^{-1}$. *When she tried playing it indoors at* 20 °C, *its lowes one had frequency* 349 Hz *and the flute had length* l. *Near the fire, she placed the flute o bench. Assuming that the period of sound waves forming each tone is proportional to th ength of the flute, determine the frequency of its lowest tone, if the flute got heated up to* 25 °C *n the meantime.*

Meggy was creating problems during a Fykos camp

Problem FoL.6 ... little cubes

Assume the molecules of water to be cubes in close contact (the volume of water is completel
filled by the cubes). Determine the number of molecules that fit in $V' = 1\,\mathrm{nm}^3$ *of volume*
Consider the water to be at standard conditions ($T \approx 300\,\mathrm{K}$, $p \approx 1\,\mathrm{atm}$*), look up the necessar*
values of water density, molar mass and Avogadro's number (if you don't know them by heart)

Mirek didn't see the point of including such primitive problems in master's studies

Problem FoL.7 ... methane

At the bottom of, for example, the South China Sea, is a substance know as methane clathrate. It's a crystal of water ice with gaps in which methane molecules are enclosed. These gaps are present in common ice as well, but they aren't filled by any substance. The chemical formula of this clathrate is $CH_4 \cdot 5.75H_2O$. *How many* m^3 *of methane can be released to the atmosphere at temperature* 0 °C *and pressure* 100 kPa *from* $1\,m^3$ *of the clathrate? Water ice density is* $917\,kg \cdot m^{-3}$.

Sweet memories of Katka's high school years

Problem FoL.8 ... do it tomorrow

Compute how long would day on Earth have to be in order to cause weightlessness at th equator. The Earth's radius is $R = 6,400\,\mathrm{km}$, *its mass is* $M = 6 \cdot 10^{24}\,\mathrm{kg}$, *the gravitationa constant is* $G = 6.7 \cdot 10^{-11}\,\mathrm{kg}^{-1} \cdot \mathrm{m}^3 \cdot \mathrm{s}^{-2}$. *Assume the Earth to be spherical.*

Mirek was sitting by the window watching time fly

Problem FoL.9 ... Toricelli reloaded

We are on a space station with standard atmospheric pressure $p_\mathrm{a} = 101\,\mathrm{kPa}$ *inside, performin*
he Toricelli experiment. We take a bottle of mercury, stick a long test tube inside and when it i
ull, place it vertically in such a way that the open end is submerged under the surface of mercur
n the bottle. During the experiment, the space station is sufficiently far from the gravitationa
nfluence of any celestial bodies and is accelerating in the direction of the closed end of the tes
ube (after the above described manipulation) with acceleration $a = 20\,\mathrm{m{\cdot}s^{-2}}$. *How high wi*
he mercury column in the test tube be? The density of mercury is $\varrho = 13\,600\,\mathrm{kg{\cdot}m^{-3}}$.

Mirek was disappointed by an easy lecture, so he upgraded the problems

Problem FoL.10 ... effectiveness evaluation

Karel and Aleš have one homogeneous brick each. The bricks have cuboid shapes with edg lengths $a \times a \times 2a$, where $a = 5\,\mathrm{cm}$. The mass of each brick is $m = 1\,\mathrm{kg}$. Karel puts his bric down and starts flipping it in one direction always about the shorter edge (looking from th side, the longer edge and the shorter edge alternate in touching the ground), until the bric crosses a distance $d = 150\,\mathrm{cm}$. Aleš is also flipping his brick until it crosses the distance d, bu he is flipping it about the longer edges only (only the square base of the cuboid is ever visibl from the side). The bricks do not slip and collisions with the ground are inelastic. How muc more work will Karel perform? The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Mirek was evaluating the others' performances

Problem FoL.11 ... 'murican

Náry does not like unpleasant surprises. How far from the muzzle of an air gun does he hav
o stand in order to hear the shot exactly one second before he would get hit? The speed o
ound is $334\,\mathrm{m{\cdot}s^{-1}}$*, the exit velocity of the bullet is* $152\,\mathrm{m{\cdot}s^{-1}}$*, the elevation angle of the sho*
s $10\circ$*. Neglect air resistance. The term "heard" is considered equivalent to Náry being reache*
y the sound wave. The time the bullet spends inside the muzzle is also negligible. Comput
he answer to metre precision. *Kiki would shoo*

Problem FoL.12 ... from Prague to Brno

You are driving from Prague to Brno with velocity $v = 150\,\mathrm{km{\cdot}h^{-1}}$ *on the D1 motorway. Th*
ength of this path is $d = 196\,\mathrm{km}$. *How many times will you pass by a RegioJet bus (also headin*
o Brno) during driving, if you know that every half hour, a bus leaves Prague for Brno wit
elocity $v_\mathrm{b} = 80\,\mathrm{km{\cdot}h^{-1}}$ *and you leave Prague at the same time as one bus? (Don't count tha*
us, However, if you meet a bus when arriving at Brno, do count that one.) Assume that ther
re no traffic jams or closed sections of the motorway due to construction.

Dominika had an idea when travelling back from a FYKOS camp

Problem FoL.13 ... and the Sun in Ophiuchus means that...

What is the maximum force that Jupiter can exert on a FYKOS member on Earth? The FYKO member has mass $m_{\mathrm{F}} = 85.6\,\mathrm{kg}$*, look up the remaining parameters. Assume that the orbit of Jupiter and Earth are circular with radii equal to the lengths of their semi-major axes an coplanar.* *Karel heard that this is good argument for weak believers in astrology*

Problem FoL.14 ... bounce from zero

In the xy plane, there is a perfectly reflecting mirror represented by the curve $x^2 + y^2 = 1$ < 0. We send a light ray towards the concave side of the mirror along the line $x = \sqrt{3}/2$ How many times will the ray reflect before flying back to the half-plane $y > 0$?

Mirek was too lazy to turn

Problem FoL.15 ... St. Elmo's sphere

What's the minimum radius r of a conducting sphere such that there's no spontaneous discharge if the charge of the sphere is $Q = 1\,\mathrm{C}$? In air, spontaneous discharge occurs if the electric field reaches $E_{\max} = 25\,\mathrm{kV{\cdot}cm^{-1}}$.

Erik likes easy problems

Problem FoL.16 ... under pressure

We put a metal rod of length $l = 1\,\mathrm{m}$ into a hydraulic press. The ends of the rod are held i
way that creates a small stress σ_0. The rod's thermal expansion coefficient is $\alpha = 1.1 \cdot 10^{-5}\,\mathrm{K}^-$
nd its Young modulus is $E = 200\,\mathrm{GPa}$. We heat up the rod by $\Delta T = 1\,\mathrm{K}$. Calculate the chang
n stress in the rod if the rams of the press do not move and are not thermally conducting.

Mirek was watching a hydraulic press

Problem FoL.17 ... long drop

We throw a polystyrene ball of radius $r = 1.0\,\mathrm{cm}$ *and density* $\varrho = 0.06\,\mathrm{g{\cdot}cm^{-3}}$ *vertically dow*
nto a ventillation shaft. In the shaft, hot warm air with density $\varrho_\mathrm{a} = 1.1\,\mathrm{kg{\cdot}m^{-3}}$ *flows up wit*
velocity $v_0 = 10\,\mathrm{m{\cdot}s^{-1}}$*. Determine the magnitude of terminal velocity of the ball with respec*
o the shaft, if the drag coefficient of the ball is $C = 0.50$ *and the acceleration due to gravit*
s $g = 9.8\,\mathrm{m{\cdot}s^{-2}}$*.* *Mirek couldn't work during backup, so he was creating problems*

Problem FoL.18 ... microrotation

The molar mass of water is $M = 18\,\mathrm{g{\cdot}mol^{-1}}$. *Its molecules have a certain rotational energ*
with characteristic scale $\varepsilon = 650\,\mathrm{J{\cdot}mol^{-1}}$. *The linear dimension of one molecule (its diameter*
s $l = 310\,\mathrm{pm}$. *Using dimensional analysis, determine the characteristic scale for the period o*
otation. The answer is the base-10 logarithm of this period in seconds.

Mirek didn't see the point of including such primitive problems in master's studies

Problem FoL.19 ... perfectly balanced

Consider a thin homogeneous disk of radius r, supported in the middle. Initially, there wer
two point masses m placed on the perimeter of the disk opposite each other so that the systen
remained balanced. We want to replace one of the masses by a pair of masses $3m/5$ and $4m/5$
both placed on the perimeter of the disk, so that the system still remained balanced. How fa

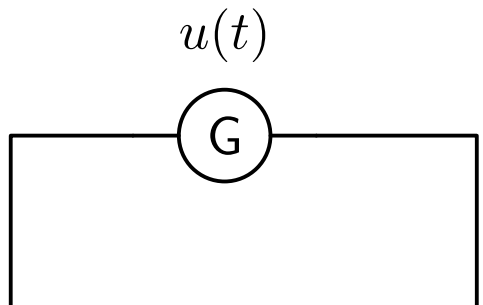
in multiples of r) from each other do the two replacement masses have to be?

Meggy was bored during an algebra lecture

Problem FoL.20 ... sine

We connect a tone generator producing a sine wave voltage $u(t)$ into a series RC circuit. The resistor in the circuit has resistance $R = 200\,\Omega$, the capacitor has capacity $C = 50\,\mu\mathrm{F}$, the maximum voltage of the source is $U = 5\,\mathrm{V}$. What's the mean charge on the capacitor?

Kuba simplified his first circuit.



Problem FoL.21 ... chain reaction

We placed a large number of balls of radius $r = 1\,\mathrm{cm}$ in a line. The distance of centers of an
wo successive balls is $d = 5\,\mathrm{cm}$. Each ball has mass $m = 20\,\mathrm{g}$ and rolling friction coefficien
$= 0.6\,\mathrm{mm}$. The first ball in the line is given a velocity $v = 1\,\mathrm{m{\cdot}s^{-1}}$ in order to start a serie
f central collisions of the balls. Which ball will be the last one to move? All collisions ar
erfectly elastic and the balls don't slip. The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$

Mirek was strolling through the St. Wenceslas square

Problem FoL.22 ... ready steady

We placed two identically charged particles with masses $m = 1\,\mathrm{g}$ and charges $q = 1\,\mu\mathrm{C}$ a height $h = 2\,\mathrm{m}$ above the ground. Afterwards, we release them and measure the velocity o one of the particles just before colliding with the ground to be $v = 10\,\mathrm{m{\cdot}s^{-1}}$. How much did th electrostatic potential energy decrease between the start of the movement and this moment The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Mirek was not so fondly remembering children's games

Problem FoL.23 ... it hangs – it hangs

In a laboratory, a polystyrene ball of mass $m_1 = 1\,\mathrm{kg}$ *and density* $\varrho = 1\,020\,\mathrm{kg{\cdot}m^{-3}}$ *hangs o*
a massless rope. The molar heat capacity of polystyrene is $c = 1\,400\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$*, its therma*
expansion coefficient is $\alpha = 7 \cdot 10^{-5}\,\mathrm{K^{-1}}$*. On another rope, we hang another ball made of th*
same material and with mass $m_2 = 0.5\,\mathrm{kg}$*. To each ball, we transfer heat* $Q = 5\,\mathrm{kJ}$*, whic*
increases the energy of the first ball by E_1 *and of the second ball by* E_2*. Find the differenc*

$E_2 - E_1$. *The acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

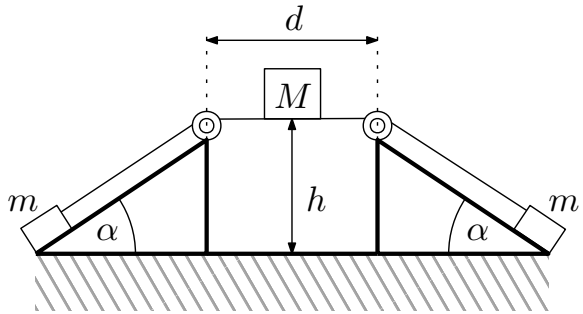
Mirek was pondering textbook problems

Problem FoL.24 ... eccentric travelling companion

Given the ratio of velocities of a planet on an eliptical trajectory around a star in the pericente (the point of the trajectory closest to the star) and in the apocenter (the point farthest from the star) $v_\mathrm{p}/v_\mathrm{a} = K = \sqrt{2}$, what's the numerical (relative) eccentricity ε of that trajectory?

Karel likes problems that deal with Kepler's 2nd law

Problem FoL.25 ... cuboids up in the air



n the figure, you see a large cuboid of mass $M = 1\,\mathrm{kg}$, pulling wo smaller cuboids of masses $m = M/4$ through pulleys along nclined planes with inclination angles $\alpha = 35°$. Before the small uboids reach the pulleys, the large cuboid will collide with the round inelastically. What is the maximum height the small uboids reach? We know the lengths marked in the figure: d = = $0.8\,\mathrm{m}$, $h = 0.6\,\mathrm{m}$; neglect the sizes of the cuboids and pulleys. Both the dynamic and stati riction coefficients between the cuboids and the inclined plane are $f = 0.5$. We measure th eight we are asking for with respect to the initial heights of the cuboids, the acceleration du o gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$. There is no friction in the pulleys.

Mirek's ingenious construction

Problem FoL.26 ... tuning

In the middle of a field, there's Pepa with a tuning fork emitting a sound with frequenc $f_1 = 440\,\mathrm{Hz}$. *You're driving away from Pepa with velocity* $v = 70\,\mathrm{km{\cdot}h^{-1}}$. *What will be th frequency* f *of the sound you hear? The speed of sound in air is* $v_\mathrm{s} = 330\,\mathrm{m{\cdot}s^{-1}}$.

Olda made it

Problem FoL.27 ... aloha snackbar

The US President's airplane has range approximately 12 600 km. *Once upon a time, it wa hijacked by terrorists shortly after takeoff, with the president on board, and vanished from th adars. A thorough search of both land and ocean was launched immediately. What area (i* km^2*) has to be searched if the terrorists can't refuel anywhere? Mikulas was kidnapped*

Problem FoL.28 ... burst

nto how many smaller droplets could a larger water droplet break, if it fell from a $h = 50\,\mathrm{cm}$ eight? Consider all droplets to be spherical. The radius of the original droplet is $r = 1\,\mathrm{cm}$ with its center of mass at height h) and the droplets it breaks into should have equal sizes The surface tension of water is $\sigma = 73\,\mathrm{mN{\cdot}m^{-1}}$ and its density is $\varrho = 1.00\,\mathrm{g{\cdot}cm^{-3}}$. We're onl ooking for an estimate of an upper bound of the number of droplets based on the energeti balance at the beginning and at the end. *Karel was watching a falling droplet*

Problem FoL.29 ... damped oscillation

There's a mass point that oscillates along the y axis. The oscillation is described by the equa
ion $y = y_\mathrm{m} \mathrm{e}^{-bt} \sin(\omega t + \varphi_0)$, *where* y_m *is the initial amplitude,* $b = 0.05\,\mathrm{s}^{-1}$ *is the dampin*
oefficient, $\omega = 200\,\mathrm{Hz}$ *is the angular frequency and* φ_0 *is the initial angular displacement*
When will the maximal amplitude decrease below $y_\mathrm{m}/2$*?*

Karel was watching the oscillation of a strongly damped spring

Problem FoL.30 ... don't stick your fingers in!

A large ceiling fan may be approximated as a disk of radius $r = 10\,\mathrm{cm}$ *and mass* $M = 0.5\,\mathrm{kg}$ *with three rods attached, pointing away from the centre of the disk. Each rod has length* $l = 40\,\mathrm{cm}$ *and mass* $m = 0.2\,\mathrm{kg}$. *The fan rotates with frequency* $f = 1\,\mathrm{Hz}$. *What work do we need to perform to stop the fan? All parts of the fan are homogeneous, neglect any friction.*

Mirek was tickling the blades

Problem FoL.31 ... dark planet

What's the radius of a sphere with Earth-like density $\varrho = 5.50\,\mathrm{g{\cdot}cm^{-3}}$*, but escape velocit*
equal to the speed of light? We're interested in the first-order approximation, i.e. neglectin
relativistic effects. Assume that the planet's density is constant (the planet is homogeneous)
Give the result as a multiple of the radius of Earth $R_\mathrm{Z} = 6,371\,\mathrm{km}$.
Karel likes to talk about relativistic effects, but doesn't like actually solving relativistic problems

Problem FoL.32 ... crash

The Moon is an approximately homogeneous sphere of mass $M = 7.3 \cdot 10^{22}\,\mathrm{kg}$ *and radius* $R =$ $= 1,700\,\mathrm{km}$, *rotating with a period* $T = 27\,\mathrm{d}$. *A megameteorite of mass* $m = M/1\,000$ *collide with the Moon's equator under an incidence angle* $\vartheta = 45^\circ$ *into the lunar soil, against th direction of rotation. The impact velocity is* $v = 10\,\mathrm{km{\cdot}s^{-1}}$. *How long will the lunar day b after the impact, if the debris from the meteorite spreads evenly over the surface of the Moon The meteorite and the Moon are both made of the same material. Neglect mutual gravitationa influence of the bodies before the impact.* Give the result as a positive number in the units c (Earth's) days.

Mirek wanted the Moon to have a moon

Problem FoL.33 ... just a little closer

There are two hooks on two opposite walls with distance $d = 1\,\mathrm{m}$. On one hook, there's a charg
$= 1\,\mu\mathrm{C}$. On the other hook, there's no charge, but a thin non-conducting massless rope o
ength $l = d$, with a ball with mass $m = 2\,\mathrm{g}$ and charge $-q$ on its other end. Find the smalles
ngle (in $^\circ$) between the rope and the wall such that the ball is at rest.

Mirek was reaching for his notebook

Problem FoL.34 ... chain reaction II

We placed a large number of identical ice cuboids with square bases of side $a = 2\,\mathrm{cm}$ in a lin
on an ice rink. The distance between centers of any two successive cuboids is $d = 5\,\mathrm{cm}$. Eac
uboid has mass $m = 20\,\mathrm{g}$ and the friction coefficients (both static and dynamic) are $f = 0.03$
The first cuboid in the line is given a velocity $v_0 = 1\,\mathrm{m{\cdot}s^{-1}}$ in order to start a series of centra
ollisions of the cuboids. Which cuboid will be the last one to move? All collisions are perfectl
nelastic, the cuboids don't tilt and are placed in such a way that they have parallel sides. Th
cceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Mirek was strolling through the St. Wenceslas square again

Problem FoL.35 ... Newton's rings

Olda was watching Newton's glass. Compute the distance of the 2nd minimum of the interferenc circles from the center, if we know that the light incident on it has wavelength $\lambda = 500\,\mathrm{nm}$ the refraction index of glass is $n_\mathrm{s} = 1.5$ and the refraction index of air is $n_\mathrm{v} = 1$. The radius c curvature of the upper glass is $r = 20\,\mathrm{cm}$. *Olda was looking at Newton's ring*

Problem FoL.36 ... short circuit

There's a simple electric circuit with two elements: a DC source with voltage $U = 1\,\mathrm{V}$ *an internal resistance* $R = 1\,\Omega$, *and a coil with* $N = 200$ *turns of copper winding (circular cross ection), length* $l = 30\,\mathrm{cm}$, *inner radius* $r_1 = 1.90\,\mathrm{cm}$ *and outer radius* $r_2 = 2.05\,\mathrm{cm}$. *What's th maximum current passing through the coil after we switch on the source? The temperature i the room is* $20\,^\circ\mathrm{C}$, *neglect any heating of the winding.*

Punchline redacted to avoid people whining about sexism

Problem FoL.37 ... exoplanet

An exoplanet of radius $r_\mathrm{p} = 8.0 \cdot 10^3\,\mathrm{km}$ *is in circular orbit around a distant star with mass* $M =$
$= 1.0 \cdot 10^{30}\,\mathrm{kg}$. *The orbit and the observer are coplanar. A transit of the planet across the sta*

was observed. The time from the first decrease in brightness until brightness returned to it
nitial value was $T = 2.0\,\text{hours}$ *and the maximum decrease in brightness was* $0.3\,\%$*. Determin*
he orbital radius. *Filip read about newly discovered exoplanets*

Problem FoL.38 ... tse-tse

Little rabbit has a prescription for sleeping pills. When he takes 1/3 *of a pill, he sleeps for hours. When he takes* 2/3 *of a pill, he sleeps for 9 hours. Once, little rabbit went nuts and took 100 pills. How many hours did he sleep, if we know that little rabbit sleeps only when the medicine's concentration in blood exceeds a certain value? Assume that the rate at which the medicine is broken down is proportional to its concentration. Neglect any medicine left over in blood from the previous days or little rabbit's death.* *Mikulas couldn't sleep*

Problem FoL.39 ... throwing peas at the wall

Mirek was sitting on an office chair with wheels at work, looking at a wall deep in thoughts n order to do something more useful, he took a sack of peas and started throwing them a wall. There are $n = 3,600$ peas in the sack, each of them weighs $m_\mathrm{h} = 0.2\,\mathrm{g}$ and Mirek i hrowing them with a frequency $f = 0.5\,\mathrm{s}^{-1}$. The peas are thrown horizontally with velocit $_\mathrm{h} = 10\,\mathrm{m{\cdot}s}^{-1}$. What will Mirek's velocity be after emptying the sack? The mass of Mirek - hair is $m_\mathrm{M} = 60\,\mathrm{kg}$, the chair moves without friction. *Mirek needed to move on*

Problem FoL.40 ... wire

There's a thin, perfectly black wire with circular cross-section of radius $r = 0.3\,\mathrm{mm}$ *in therma equilibrium at room temperature* ($T_0 = 20\,^\circ\mathrm{C}$) *in vacuum. Under these conditions, the resistanc of the wire per unit length is* $R_0 = 5\,\Omega/\mathrm{m}$. *If the thermal coefficient of resistance is* $\alpha = 0.004\,\mathrm{K}^{-1}$, *determine the current that must pass through the wire so that its temperatur stabilised at* $220\,^\circ\mathrm{C}$.

Filip burnt his finger

Problem FoL.41 ... hydrophobe

There's a water droplet on the horizontal part of an umbrella. The contact angle between th droplet and the umbrella is quite large, $\vartheta_c = 120°$*. The surface tension between water and ai is* $\sigma_{lg} = 73\,\mathrm{mN{\cdot}m^{-1}}$*. We add a bit of detergent to the droplet, decreasing the surface tensio to* $\sigma'_{lg} = 56\,\mathrm{mN{\cdot}m^{-1}}$ *and the contact angle to* $\vartheta'_c = 90°$*. How much did the surface tensio between the surface of the umbrella and the liquid have to change?* Compute the result in unit $\mathrm{mN{\cdot}m^{-1}}$!

Mirek was inspecting his broken umbrella

Problem FoL.42 ... distributed

When describing plasma in magnetic field, it's often useful to work in cylindrical coordinate elated to magnetic field lines. In these coordinates, we can model an anisotropic velocit distribution using a so-called bi-Maxwell distribution function

$$f(v_\perp, v_\parallel) = \exp\left(-\frac{\left(v_\parallel - \mu_\parallel\right)^2}{2\sigma_\parallel^2}\right) \exp\left(-\frac{\left(v_\perp - \mu_\perp\right)^2}{2\sigma_\perp^2}\right) \frac{1}{(2\pi)^{3/2}\sigma_\perp\sigma_\parallel}\,.$$

The indices $\parallel$ and $\perp$ denote components parallel to the magnetic field and perpendicular t t; the angular component has been removed by integration (due to axial symmetry). Th parameters of the velocity distribution are known to be $\sigma_\parallel = 10^6\,\mathrm{m{\cdot}s^{-1}}$, $\sigma_\perp = 4\cdot10^5\,\mathrm{m{\cdot}s^{-1}}$ an $\mu_\parallel = 4\cdot10^4\,\mathrm{m{\cdot}s^{-1}}$, $\mu_\perp = 3\cdot10^4\,\mathrm{m{\cdot}s^{-1}}$. Determine the mean velocity of a particle described b he distribution $f(v_\perp, v_\parallel)$.

Mirek submitted a practical problem

Problem FoL.43 ... merry-go-round

There's a box of mass $m = 5\,\mathrm{kg}$ and negligible size standing on a merry-go-round. The bo
s $r = 10\,\mathrm{m}$ far from the center of the merry-go-round; the static friction coefficient betwee
hem is $f_0 = 1$. The merry-go-round starts rotating from rest with linearly increasing angula
cceleration $\varepsilon = kt$, where $k = 1\,\mathrm{s}^{-3}$. How long since the start of the rotation will it take for th
ox to fly away from its initial position? Kuba wanted a problem where you need Coriolis..

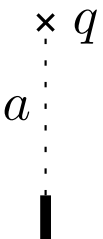
Problem FoL.44 ... twin paradox

One of two twins leaves in a rocket for a planet $s = 5\,\mathrm{ly}$ far from the Earth, while the other twi stays at their home planet. The rocket moves with velocity $v = 0.2c$. At half the distance s, th cosmonaut twin enters a fast module, which starts moving with respect to the original rocke with velocity $u = 0.1c$ towards the original goal of the rocket. How many years younger wi the cosmonaut twin be compared to the twin at the home planet? Neglect any effects cause by acceleration or deceleration of the space vessels (as if they only passed by each other an gravity did not exist). In order to avoid trouble with relativity of simultaneity, compare th times spent by each twin waiting for the arrival of the fast module at the goal.

All special relativity problems are washed ou

Problem FoL.45 ... vampirectrical

There's a charged non-conducting cross and a point charge, placed as shown in the figure. We know $a = 20\,\mathrm{cm}$, $q = 1\,\mu\mathrm{C}$ *and the linear charge density of the cross* $\lambda = q/a$. *What's the magnitude of the electrostatic force acting on the charge? Mirek was defending himself from diabolical problems.*



Problem FoL.46 ... focus

On one side of a thin biconvex lens with identical radii of curvature $R = 20\,\mathrm{cm}$ *is a spherica mirror placed coaxially in such a way that a ray that passes through the lens is reflected bac towards the lens. What is the (positive) focal length of such a system? The refractive index o glass is* $n = 1.5$.

Damn optics again

Problem FoL.47 ... Sisyphean labour

Sisyphos is standing at the bottom of a hemispherical hole of radius $R = 100\,\mathrm{m}$. *His task is t*
push an approximately spherical, bumpy rock of mass $M = 1\,000\,\mathrm{kg}$ *and radius* $r = 50\,\mathrm{cm}$. *Th*
effective rolling friction coefficient of the rock is $\xi = 4\,\mathrm{cm}$. *Sisyphos is pushing the rock up wit*
negligible constant velocity while exerting a force F_s, *horizontally towards the center of mass o*
the rock. Sisyphos runs out of stamina after performing work $W_\mathrm{s} = 200\,\mathrm{kJ}$. *By how much wi*
he lift the center of mass of the rock before it happens? Neglect changes in potential energy o
Sisyphos, the acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Mirek tried to resolve all unread e-mails

Problem FoL.48 ... spin it

There is a bicycle on a bicycle stand. We know that its rear wheel's moment of inertia aroun
ts axis is $I = 0.5\,\mathrm{kg{\cdot}m^2}$ *and its radius is* $r = 30\,\mathrm{cm}$. *We speed up the wheel by repeatedl*
itting it with a hand tangentially on its circumference in the direction of its rotation. Th
peed of each hit in the rest reference frame is $w = 5\,\mathrm{m{\cdot}s^{-1}}$, *the mass of the hand is* $m = 1.5\,\mathrm{kg}$
he hand loses all its velocity in the reference frame of the point of impact after the impac
the hand is moving around an elbow, but we neglect rotation of the forearm). What will th
elocity of a point on the perimeter of the wheel after 10 hits be?

Mirek tampered with something again

Problem FoL.49 ... all day long

Imagine that the observable universe is spherical with diameter $d = 10^{27}$ m *and filled with star*
placed in the lattice points of a cubic grid with side length of the unit cell $a = 10^{19}$ m. *The star*
are identical and their radiation is isotropic with luminosity $L = 10^{27}$ W *and radius* $r = 10^{9}$ m
The stars don't block other stars' light; there is no absorption. What's the incident power pe
square metre near the centre of the universe in the middle of a body diagonal of a unit cell
Compute the base-10 logarithm of this value! *Mirek was trying to see through a dense forest*

Problem FoL.50 ... comfy by the pond

t's summer and we're resting on a bank of a deep pond. The sun is shining, the acceleration du
o gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*, the surface tension of water is* $\sigma = 72.8\,\mathrm{mN{\cdot}m^{-1}}$*, water density i*
$= 1\,000\,\mathrm{kg{\cdot}m^{-3}}$*. At a distance* $d = 30.0\,\mathrm{m}$ *from the bank, a backswimmer appears and create*
ircular waves on the surface. When we focus on one wave, we can see that it's moving wit
peed $c = 1.00\,\mathrm{m{\cdot}s^{-1}}$*. How long will it take for the wavefront of the backswimmer's waves t*
each the bank? You may use the formula for phase velocity of deep-water waves $c = \sqrt{\frac{g}{k} + \frac{\sigma k}{\varrho}}$
where k *is the wave number.* *Mirek confused gravity waves for gravitational waves*

Problem FoL.51 ... toast

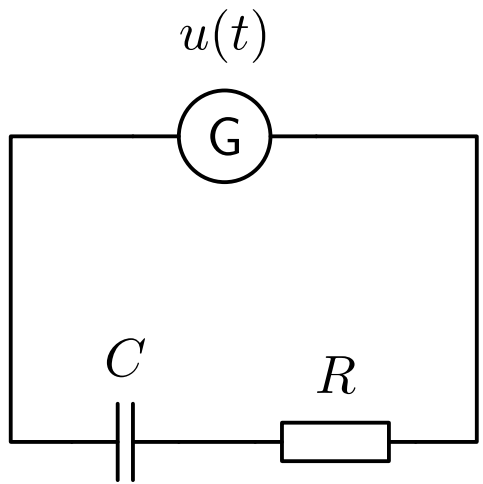
We'd like to make a toast using a very strange glass. It's a hollow cube of inner volume $V = 1$ l which is standing on one corner (a body diagonal is vertical). Through a small opening nea the top vertex, we fill the cube with $V' = 1/31$ of wine. How high above the ground (i.e. th vertex the cube is standing on) will the wine surface be? Neglect the thickness of the cube' ides. Compute the result as a multiple of the cube's edge length!

Mirek remembered stereometry and had mixed feelings

Problem FoL.52 ... saw

We connect a tone generator producing a sawtooth wave voltage $u(t)$ nto a series RC circuit. The resistor in the circuit has resistance $R = 200\,\Omega$, the capacitor has capacity $C = 50\,\mu\mathrm{F}$, the maximum voltage of the source is $U = 5\,\mathrm{V}$. For the period of voltage $T = = RC$, calculate the current passing through the circuit when the voltage of the source is at a maximum (specifically, instantly after the source voltage reaches the maximum) a long time after the generator s switched on.

Kuba thought there weren't any decent circuit problems yet.



Problem FoL.53 ... you shall not transmit!

Consider a simple model of a static one-dimensional atmosphere, denoting its relevant coordi
nate (elevation) by h. If we took a thin slice of this atmosphere with thickness Δh, its reflectanc
would be $r(h)\Delta h$. What fraction of incident light is transmitted by the atmosphere if $r(h) =$
$= r_0 \mathrm{e}^{-kh}$, where $k = r_0 = 0.1\,\mathrm{m}^{-1}$? *Kuba, an old organiser, decided to make a problem*

Problem FoL.54 ... transit

Consider a system in which infinitely many discrete states exist, indexed by an index $i \in \mathbb{N}$ (positive integers). The state of the whole system can be described by a state vector $\boldsymbol{f} = (f_1, f_2, \dots)$, where f_i denotes the number of objects in state i. The time evolution of the system occurs in discrete steps. The probability of transition of an object from a state i to the state $i+1$ in one step is $P_{i,i+1} = 1/i^2$, the probability of transition from a state i to the state $i-1$ is $P_{i,i-1} = 1 - P_{i,i+1}$. It follows from basic properties of probability that transitions between non-adjacent states or remaining in the same state isn't allowed. Initially, all objects are in state f_1. Find the stationary state the system converges to and compute the number of objects in the first state as a multiple of the total number of objects in the system N. Assume that N is a very large number. The total number of objects remains constant during the time evolution.

Mirek was transitioning from the dorm to school

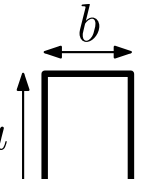
Problem FoL.55 ... megamagnet

There's a cylindrical coil of length $l = 100\,\mathrm{km}$ *and base diameter* $d = 1\,\mathrm{km}$ *with* $N = 10$ *turns. There's a current* $I = 1\,\mathrm{kA}$ *passing through the coil. Determine the electric field fel when flying out of the coil parallel to its axis at a distance* $r = 1\,\mathrm{m}$ *from that axis with velocit* $= 1,000\,\mathrm{km{\cdot}s^{-1}}$. *Neglect any effects of general relativity.* *Xellos was watching Red Dwar*

Problem M.1 ... tower builder

We have glued together a tower from three identical blocks with edge lengths a, b (see the picture). Find the largest possible ratio a/b for which the tower would still be stable.

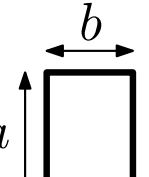
Náry and Lego.



Problem M.2 ... tower builder returns

We have built a tower from three identical blocks with edge lengths a, b. Find the largest possible ratio a/b for which the whole tower would not collapse immediately.

Náry and Lego.



Problem M.3 ... collision imminent

Calculate the lowest possible energy loss resulting from a perfectly inelastic central collision o
two rigid balls with masses $m = 3\,\mathrm{kg}$ and $M = 2\,\mathrm{kg}$. The speeds before the collision are $5\,\mathrm{m{\cdot}s^{-1}}$
and $6\,\mathrm{m{\cdot}s^{-1}}$ (with respect to a static observer). *Náry, life experience*

Problem M.4 ... stick like no other

Rigid homogeneous stick of length $l = 1\,\mathrm{m}$ has been thrown upwards in a uniform gravitationa
field. During its flight, there was a moment when the translational kinetic energy equaled th
rotational kinetic energy. We know that in that precise moment the speed of the stick's cente
of mass was $9\,\mathrm{m{\cdot}s^{-1}}$. Find the instantaneous angular velocity of the stick upon its release.

Everybody once met a rotating stick

Problem E.1 ... charged Fykosák

Electrically neutral Fykos bird has been endowed with 10^{12} *electrons. It then flies into a ho*
mogeneous electric field with speed $v_0 = 3.6\,\mathrm{km{\cdot}h^{-1}}$ *in a direction opposite to the directio*
of the field lines. Electric intensity of the field is $500\,\mathrm{V{\cdot}m^{-1}}$*. What will be the speed* v *of th*
Fykos bird after $100\,\mathrm{m}$ *long flight in the electric field? Consider a point-like bird with mass* m =
$= 0.5\,\mathrm{kg}$.

Faleš has been watching birds at a bus stop. .

Problem E.2 ... Náry's hoverboard

How much current would have to flow through a horizontal wire $1\,\mathrm{m}$ *long to make the wir*
evitate above the ground? Magnetic induction is $5 \cdot 10^{-5}\,\mathrm{T}$ *at the place where the wire is an*
points to the north in the horizontal plane. The wire has a mass $50\,\mathrm{g}$*. The direction of the wir*
s from west to east.

Faleš and Náry playing with a compass

Problem E.3 ... astonished Náry

Náry was playing with electrical components and ended up with four peculiar things. He wa
old by others that they are a resistor with resistance $R = 10\,000\,\Omega$*, capacitor with capacit*
$C = 1\,\mu\mathrm{F}$*, inductor with inductance* $L = 10\,\mathrm{H}$ *and alternating current source with voltag*
$U = 230\,\mathrm{V}$ *and frequency* $f = 50\,\mathrm{Hz}$*. By some strange coincidence he managed to build a serie*
ircuit. Knowing what he is playing with, he immediately computed the absolute value of phas
hift between current and voltage in the circuit. What number did he get?

Faleš and Náry during experimental afternoon

Problem E.4 ... it's freezing

An arc lamp has resistivity $0.2\,\Omega$ *and it is connected to voltage* $U = 60\,\mathrm{V}$*. How much heat doe*
t release during $1\,\mathrm{min}$*? Give the result in* MJ. *Faleš shivered with cold*

Problem X.1 ... Odysea and Prometheus

There are two space-ships, Odysea and Prometheus. Both of them have rest length L_0 and both are approaching the Earth, but from opposite direction. They meet in a point where Odysea has speed $0.25c$ and Prometheus has speed $0.75c$. How long do they appear to each other in the moment when they pass each other? Determine the length as a multiple of L_0.

Faleš has been watching Stargate

Problem X.2 ... who's gonna survive

Let us have a beam of neutrons with kinetic energy $T = 10\,\mathrm{keV}$*. Mean lifetime of a neutron i* $\tau_\mathrm{n} = 925\,\mathrm{s}$ *and its rest energy is* $m_\mathrm{n}c^2 = 939.6\,\mathrm{MeV}$*. What percentage of neutrons in the bean decays during passage through distance* $l = 10\,\mathrm{m}$*?* *Faleš was watching Independence day*

Problem X.3 ... the winter, again

Consider a block of ice with mass $m_{\mathrm{i}} = 5\,\mathrm{kg}$ *and temperature* $t_{\mathrm{i}} = -2\,^{\circ}\mathrm{C}$. *We place* $m_{\mathrm{Fe}} = 3\,\mathrm{k}$ *block of iron heated to* $t_{\mathrm{Fe}} = 1\,333\,^{\circ}\mathrm{C}$ *on the block of ice. Determine the resulting temperatur of the system. The specific heat capacity of ice is* $c_{\mathrm{i}} = 2.1\,\mathrm{kJ{\cdot}kg^{-1}{\cdot}K^{-1}}$, *the heat of fusion of ic is* $l_{\mathrm{im}} = 334\,\mathrm{kJ{\cdot}kg^{-1}}$, *the specific heat capacity of water is* $c_{\mathrm{w}} = 4.18\,\mathrm{kJ{\cdot}kg^{-1}{\cdot}K^{-1}}$, *the laten heat of evaporation of water is* $l_{\mathrm{vw}} = 2.26\,\mathrm{MJ{\cdot}kg^{-1}}$ *and the specific heat capacity of iron i* $_{\mathrm{Fe}} = 473\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$

Faleš has seen ice for the first time this yea

Problem X.4 ... tiler

Are there such plane edge-to-edge tilings where, in every vertex, t equilateral triangles, s squares and h regular hexagons meet? We want to know how many such tilings exist. Numbers t, s, h are non-negative integers. Sizes of polygons of the same type in one tiling are identical. The arrangement of polygons about the vertex does not matter (different arrangements for fixed s, t, h count as one).

Mirek got a new book